

# Hamiltonian Edge Separation in Barnette Graphs: Certified Extremal Values and an Exact Double-Ladder Formula

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## Abstract

For a Hamiltonian graph  $G$ , we study the minimum number  $\text{hsep}(G)$  of Hamiltonian cycles needed to distinguish every ordered pair of distinct edges: for each  $(e, f)$ , some selected cycle must contain  $e$  and avoid  $f$ . We report an exhaustively certified finite computation over all 22,263 non-isomorphic Barnette graphs of orders at most 36. The resulting extremal values are independently checked by explicit Hamiltonian-cycle covers and matching lower certificates or complete-universe exclusion. In particular, the maximum at order 36 is 49 and is attained uniquely by the graph with canonical hash `2f96ada16c46cd2bd038b97d5af44f46ed14522cba1edf3fa041d66e02107bcc`. The unique maximizers at orders divisible by four form, over the certified range, a double-ladder family linked by a fixed four-vertex square insertion. We define this family, prove its Hamiltonian-cycle classification, and prove for all odd  $a, b \geq 3$  the exact formula

$$\text{hsep}(D(a, b)) = \frac{(a + 2)(b + 2) - 1}{2}.$$

The lower bound is a packing of ordered edge requirements whose supports are exception singletons, boundary-turn singletons, and disjoint interior dominoes. The matching upper bound selects the four connector-exception cycles, every boundary turn, and one parity class of interior turns; the resulting edge-incidence signatures form an antichain. Three values frozen before computation and later certified for  $D(9, 9)$ ,  $D(11, 9)$ , and  $D(11, 11)$  provide prospective finite confirmation. They do not prove extremality at orders 40, 44, or 48: general extremality of balanced double ladders remains open. The terminology and its relationship to prior separation notions also remain subject to a systematic literature review.

## 1 Introduction

A family of Hamiltonian cycles can be viewed as a binary measurement system on the edges of a graph: an edge receives one bit for every cycle, indicating whether the cycle contains it. Requiring every ordered pair  $(e, f)$  to be separated asks for a coordinate at which the signature of  $e$  is one and that of  $f$  is zero. This requirement is stronger than mere distinctness of the signatures and leads naturally to an antichain condition.

This manuscript develops that question for Barnette graphs, namely simple, planar, bipartite, cubic, and 3-connected graphs. Its first purpose is to state precisely what has been established by a finite computer-assisted proof. Its second purpose is to isolate a recurrent double-ladder structure among the certified extremal examples. The manuscript deliberately distinguishes four levels of evidence:

- (i) ordinary mathematical propositions proved in the text;
- (ii) finite computational results supported by explicit, independently checkable certificates;
- (iii) prospective computations whose values were frozen before evaluation; and

(iv) open extremal conjectures beyond the certified census range.

The certified computation gives all extremal values from order 8 through order 36 whenever the Barnette census is nonempty. The order-10 census is empty and odd orders are impossible for cubic graphs. At the nonempty orders, exact upper certificates are Hamiltonian edge-separating families, while lower bounds are supplied either by packing requirements or by exhaustive exclusion over a certified complete Hamiltonian-cycle universe. Importantly, no greedy cover size is used as an exact value.

The structurally recurring examples are denoted by  $D(a, b)$ . Two ladder strips are joined by four fixed terminal edges. The certified unique maximizers at orders (8,12,16,20,24,28,32,36) are respectively

$$D(1, 1), D(3, 1), D(3, 3), D(5, 3), D(5, 5), D(7, 5), D(7, 7), D(9, 7).$$

The tied extremal graphs at orders (22,26,30,34) have a different face structure and are not forced into this uniform family.

The finite pattern leads to an infinite-family theorem. A rail/rung recurrence classifies all Hamiltonian cycles of  $D(a, b)$  for odd  $a, b \geq 3$ . A symbolic packing supplies a lower bound for hsep, while a boundary-plus-parity family of Hamiltonian cycles supplies a matching upper bound through an incidence-signature antichain. Consequently,

$$\text{hsep}(D(a, b)) = \frac{(a+2)(b+2)-1}{2} \quad (a, b \geq 3 \text{ odd}).$$

This settles the invariant on the double-ladder family, not its extremality among all Barnette graphs of the same order.

**Literature-review status. TODO:** A systematic peer-reviewed literature review is required before any novelty claim is made. It must cover separating and completely separating systems, edge separation by cycles, Hamiltonian-cycle covers, Barnette graph generation and census results, and reducible configurations or expansion operations. The notation hsep and the descriptive name used here are therefore provisional.

The remainder of the paper defines the invariant and its certificate logic, describes the census and verification pipeline, states the certified extremal sequence, analyzes the double-ladder family, proves matching symbolic bounds, and separates prospective confirmations from open extremal conjectures.

## 2 Definitions and certificate principles

All graphs in this manuscript are finite and undirected. A Hamiltonian cycle is regarded as an undirected edge set; reversal and cyclic rotation do not produce new cycles.

**Definition 2.1** (Barnette graph). A *Barnette graph* is a simple, planar, bipartite, cubic, and 3-vertex-connected graph.

Let  $\mathcal{H}(G)$  denote the set of undirected Hamiltonian cycles of a Hamiltonian graph  $G$ .

**Definition 2.2** (Hamiltonian edge separation). For distinct edges  $e, f \in E(G)$ , a Hamiltonian cycle  $C$  *covers* the ordered requirement  $(e, f)$  if  $e \in C$  and  $f \notin C$ . A family  $\mathcal{C} \subseteq \mathcal{H}(G)$  is *Hamiltonian edge-separating* if it covers every ordered pair of distinct edges. Define

$$\text{hsep}(G) = \min \left\{ |\mathcal{C}| : \begin{array}{l} \mathcal{C} \subseteq \mathcal{H}(G), \\ \text{for all distinct } e, f \in E(G), \text{ some } C \in \mathcal{C} \text{ has } e \in C, f \notin C \end{array} \right\}. \quad (1)$$

**Definition 2.3** (Barnette extremal sequence). For an order at which the relevant values are defined, put

$$M_B(n) = \max\{\text{hsep}(G) : G \text{ is a Barnette graph and } |V(G)| = n\}. \quad (2)$$

If no Barnette graph of order  $n$  exists,  $M_B(n)$  is undefined. In the certified range every graph in every nonempty census is Hamiltonian, so equation (2) is unambiguous there.

Given an ordered family  $C_1, \dots, C_k$ , associate with an edge  $e$  its incidence signature

$$\sigma(e) = \{i : e \in C_i\} \subseteq \{1, \dots, k\}.$$

**Proposition 2.4** (Signature antichain). *A family  $C_1, \dots, C_k$  is Hamiltonian edge-separating if and only if the edge signatures form an antichain: for all distinct  $e, f$ , neither  $\sigma(e) \subseteq \sigma(f)$  nor  $\sigma(f) \subseteq \sigma(e)$ .*

*Proof.* The requirement  $(e, f)$  is covered exactly when some index belongs to  $\sigma(e) \setminus \sigma(f)$ . Covering both ordered requirements  $(e, f)$  and  $(f, e)$  is therefore equivalent to incomparability of the two signatures.  $\square$

In particular, if  $G$  has  $m$  edges and  $k$  selected cycles, Sperner's theorem gives the necessary inequality

$$m \leq \binom{k}{\lfloor k/2 \rfloor}.$$

This general counting obstruction is kept separate from graph-specific optimality certificates.

**Lemma 2.5** (Primal and packing certificates). *If  $k$  explicit Hamiltonian cycles cover every ordered edge-pair requirement, then  $\text{hsep}(G) \leq k$ . If there are  $k$  requirements such that every Hamiltonian cycle covers at most one of them, then  $\text{hsep}(G) \geq k$ . Consequently, matching certificates prove  $\text{hsep}(G) = k$ .*

*Proof.* The first statement is the definition. For the second, each of the  $k$  requirements must be assigned to a selected cycle that covers it, while no selected cycle can receive two requirements. Hence at least  $k$  cycles are necessary.  $\square$

When no matching packing is available, a lower bound in this project is called independently certified only if a standalone verifier checks a complete Hamiltonian-cycle universe and exhaustively excludes every cover below the claimed value.

## 3 Computational method

### 3.1 Census generation and graph validation

Complete non-isomorphic censuses were generated with plantri 5.8 (Brinkmann et al., n.d.). For target order  $n$ , the dual generation command has the form

$$\text{plantri -b -c3 -d T -}, \quad T = n/2 + 2.$$

The binary `planar_code` stream retains the rotation system. Each record was checked for the requested order and edge count, simplicity, cubicity, bipartiteness, planarity, 3-connectivity, and Euler and face arithmetic. Canonical hashes were computed from the sphere embedding; graph6, planar code, canonical edge lists, rotations, and faces were retained in the certificates. Generation

indices were required to be contiguous and canonical hashes unique. Census completeness was never inferred from a retained candidate subset.

The plantri command, version, executable digest, environment, worker count, and runtime were recorded per order. At most four workers were used. Expensive stages used deterministic ordering, atomic checkpoints, and resumable output.

### 3.2 Hamiltonian cycles and separating covers

For a cubic graph of order  $n$ , there are  $m = 3n/2$  edges and therefore  $m(m - 1)$  ordered edge-pair requirements. At order 36 this is  $54 \cdot 53 = 2,862$  requirements per graph. Every emitted cycle was checked for closure, vertex uniqueness, spanning, and edge validity before it was used as a set-cover column.

For the smaller censuses, the complete undirected Hamiltonian-cycle universe was enumerated for every graph and the set-cover problem was solved exactly. For larger censuses, the pipeline first constructed deterministic separating covers for every graph and ranked the difficult tail. Once a candidate value  $k$  had an exact lower and upper certificate, a competitor needed only a verified cover of size at most  $k$ , or at most  $k - 1$  to establish uniqueness. Complete-universe enumeration and exact optimization were then restricted to graphs that could still attain the maximum. A timeout or incomplete universe was not accepted as a conclusion.

### 3.3 Three levels of evidence

The proof pipeline deliberately separates the following objects.

**Solver output.** A proposed cycle, cover, optimum, or infeasibility result produced by a search or optimization implementation.

**Emitted certificate.** Canonically encoded graphs and explicit cycles, together with either packing requirements or a certified complete cycle universe suitable for exhaustive exclusion.

**Independent verification.** A standalone Python standard-library program reparses the artifacts, checks graph and cycle structure, evaluates every ordered requirement, checks lower certificates, order summaries, and SHA-256 manifests, and does not import the project, NetworkX, plantri, SAT, or optimization packages.

The standalone verifier checks the emitted census records and their internal consistency; it does not independently prove the enumerative completeness of plantri. The finite census-completeness claim rests on the recorded complete plantri invocation together with contiguous output and the subsequent record audits.

For every nonmaximizer the final package contains an independently verified cover strictly smaller than the maximum at that order. For every maximizer it contains an exact primal certificate and either a matching packing or a complete-universe exhaustive lower proof. Thus a greedy witness count is never reported as  $\text{hsep}(G)$ .

## 4 The certified finite extremal sequence

**Computational Result 4.1** (Certified values through order 36). The complete Plantri 5.8 censuses and independent certificate verification give the values in table 1. Across the nonempty orders, 22,263 non-isomorphic graphs were audited. Every maximum and tie count was independently verified, and every nonmaximizer has a verified separating cover strictly below the maximum of its order.

Table 1: Certified Barnette Hamiltonian edge-separation extrema. The last column is the number of undirected Hamiltonian cycles in each maximizer; within every tied order all maximizers have the displayed count.

| $n$ | graphs | $M_B(n)$ | maximizers | Hamiltonian cycles |
|-----|--------|----------|------------|--------------------|
| 8   | 1      | 6        | 1          | 6                  |
| 12  | 1      | 8        | 1          | 8                  |
| 14  | 1      | 10       | 1          | 12                 |
| 16  | 2      | 12       | 1          | 14                 |
| 18  | 2      | 10       | 1          | 16                 |
| 20  | 8      | 17       | 1          | 20                 |
| 22  | 8      | 13       | 2          | 28                 |
| 24  | 32     | 24       | 1          | 30                 |
| 26  | 57     | 18       | 5          | 40                 |
| 28  | 185    | 31       | 1          | 40                 |
| 30  | 466    | 25       | 3          | 60                 |
| 32  | 1,543  | 40       | 1          | 54                 |
| 34  | 4,583  | 32       | 7          | 80                 |
| 36  | 15,374 | 49       | 1          | 68                 |

Every cubic graph has even order, since the degree-sum identity gives  $3|V| = 2|E|$ . Hence all odd orders from 9 through 35 are impossible. The complete order-10 Plantri census is empty. These undefined entries are not optimization failures.

The order-36 statement is especially useful as a regression target:

$$M_B(36) = 49,$$

with a unique maximizer of canonical hash

$$2f96ada16c46cd2bd038b97d5af44f46ed14522cba1edf3fa041d66e02107bcc.$$

Its complete universe contains 68 undirected Hamiltonian cycles. A 49-cycle primal certificate and a 49-requirement packing certificate prove its exact value. All 15,373 competitors have independently checked covers of size at most 48.

Matching packings certify the displayed maximizers at orders 8, 12, 16, 20, 24, 28, 32, 36. At orders 14, 18, 22, 26, 30, 34, at least one displayed exact result uses exhaustive set-cover exclusion over a certified complete cycle universe instead. This distinction concerns the form of the lower certificate, not the exactness of the result.

*Status note.* Computational Result 4.1 is an exhaustive finite computational theorem about the listed orders. It is not a proof of any formula for arbitrary  $n$ , nor does it assert anything about orders 38 or 40.

## 5 The double-ladder family

Let  $L_a = P_{a+1} \square K_2$  be a ladder with vertices  $A_i^r$ , where  $0 \leq i \leq a$  and  $r \in \{0, 1\}$ . Its rungs are  $A_i^0 A_i^1$ , and its rails are  $A_i^r A_{i+1}^r$ . Define a second, disjoint ladder  $L_b$  on vertices  $B_j^r$ .

**Definition 5.1** (Double ladder). For positive integers  $a, b$ , the graph  $D(a, b)$  consists of  $L_a$  and  $L_b$  together with the four connector edges

$$A_0^0 B_0^0, \quad A_0^1 B_b^0, \quad A_a^0 B_0^1, \quad A_a^1 B_b^1. \quad (3)$$

Thus

$$|V(D(a, b))| = 2(a + b) + 4.$$

### 5.1 The square insertion

Suppose  $uvyxu$  is a facial quadrilateral, so that  $uv$  and  $xy$  are opposite edges. Replace  $uv$  by  $upqv$ , replace  $xy$  by  $xrsy$ , and add  $pr$  and  $qs$ . The four new vertices induce the facial cycle  $pqsrp$ . In a ladder cell this operation replaces one cell by three and increases the ladder length by two.

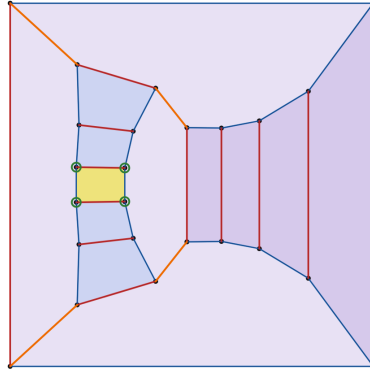


Figure 1: Annotated  $D(5, 5)$ . Blue edges are rails, red edges are rungs, orange edges are terminal attachments, and the highlighted four-vertex tile is a certified square insertion. The inverse operation deletes the tile and restores the two opposite rail edges.

**Lemma 5.2** (Preservation under facial square insertion). *Let  $G$  be a simple, planar, bipartite, cubic, 3-connected graph, and apply the preceding operation to opposite edges of a facial quadrilateral. The resulting graph  $G^+$  has the same five properties.*

*Proof.* The four terminals are distinct and the inserted vertices are new, so simplicity is immediate. Drawing the three-cell ladder inside the old face proves planarity. Each old terminal loses one edge and gains one edge; every new vertex has two neighbours on the inserted 4-cycle and one old terminal. Thus cubicity is preserved. If the facial quadrilateral is  $uvyxu$ , then  $u, y$  lie in one bipartition class and  $v, x$  in the other. The unique alternating coloring along the two new length-three paths makes both  $pr$  and  $qs$  bichromatic, proving bipartiteness.

It remains to prove 3-connectivity. Let  $S \subseteq V(G^+)$  with  $|S| \leq 2$ , and call  $p, q, r, s$  the new vertices. If  $S$  contains no new vertex, paths in the connected graph  $G - S$  lift through the two intact replacement paths; the intact inner 4-cycle is attached to every surviving terminal. If  $S$  contains exactly one new vertex, at most one old vertex is deleted. The graph obtained from  $G$  by deleting that old vertex is 2-connected and hence has no bridge. Delete, if necessary, the one

old edge whose replacement path contains the removed new vertex. The remaining old vertices are still connected, and the other three new vertices form a path attached to at least two surviving terminals. Finally, if two new vertices are deleted, no old vertex is deleted. A 3-vertex-connected graph is 3-edge-connected, so  $G - \{uv, xy\}$  is connected. Every remaining new vertex has its own terminal edge into this connected old core. Hence  $G^+ - S$  is connected in all cases, and  $G^+$  is 3-connected.  $\square$

**Theorem 5.3.** *For all odd integers  $a, b \geq 1$ , the graph  $D(a, b)$  is a Barnette graph.*

*Proof.* The graph  $D(1, 1)$  is isomorphic to the cube. Increasing either odd parameter by two is exactly the facial square insertion of theorem 5.2 in a cell of the corresponding ladder. Starting from  $D(1, 1)$ , repeated insertions therefore reach every pair of positive odd parameters while preserving all Barnette properties.  $\square$

## 5.2 Certified occurrences and face structure

**Computational Result 5.4** (Certified double-ladder recurrence). Exact graph isomorphisms identify the unique maximizers at orders divisible by four as follows:

| $n$   | 8         | 12        | 16        | 20        | 24        | 28        | 32        | 36          |
|-------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-------------|
| $G_n$ | $D(1, 1)$ | $D(3, 1)$ | $D(3, 3)$ | $D(5, 3)$ | $D(5, 5)$ | $D(7, 5)$ | $D(7, 7)$ | $D(9, 7)$ . |

All seven consecutive transitions have reconstructible expansion certificates whose reconstructed edge set is exactly the certified target and whose reduced canonical hash is exactly the certified source.

**Proposition 5.5** (Face pattern in the recursive embedding). *For odd positive  $a, b$ , the embedding obtained from the cube by the square insertions above has  $a + b$  designated ladder-cell faces, all quadrilateral, together with two faces of length  $a + 3$  and two faces of length  $b + 3$ . When  $a = 1$  or  $b = 1$ , the corresponding two cap faces are themselves quadrilateral.*

*Proof.* For  $D(1, 1)$ , the two designated cells and four cap faces are the six faces of the cube. Increasing  $a$  by two replaces one designated cell by three, adding two quadrilateral cells. The two cap boundaries running along the modified (a)-ladder each acquire two edges; the other two cap boundaries are unchanged. This is precisely the claimed update. The argument for  $b$  is identical, so induction proves the formula.  $\square$

For the certified members from  $D(3, 3)$  onward, the two sets of designated cells are exactly the two path components of the quadrilateral face-adjacency graph. At orders 8 and 12, additional quadrilateral cap faces make this face-adjacency description degenerate, although the abstract double-ladder definition remains valid.

## 6 Hamiltonian-cycle structure

The complete Hamiltonian-cycle universes of the five primary double-ladder members from orders 20 through 36 were encoded using four connector bits, rung bits, two rail strings per ladder, four-bit cell states, and adjacent-cell transitions. Complete universes and cycle counts are also certified for the three smaller, face-degenerate members, but those three were not included in the detailed tile-state artifact. The resulting finite pattern is particularly simple.

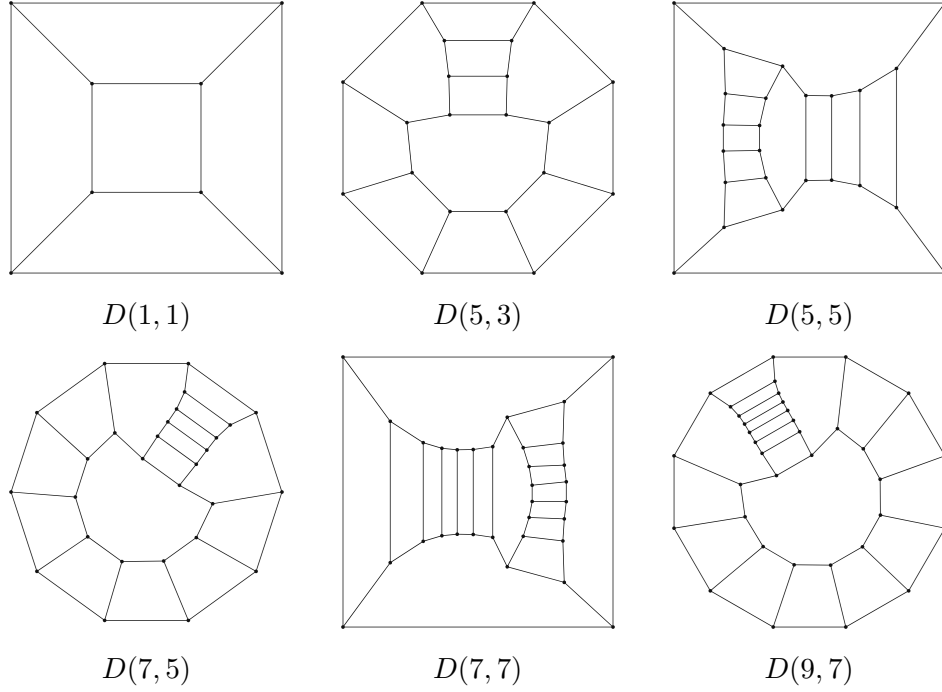


Figure 2: Selected certified double-ladder maximizers. Layouts were produced from the certified planar embeddings; geometry is used only for presentation, while all structural identifications were made from abstract edges and rotation systems.

**Computational Result 6.1** (Certified cycle counts). For each certified graph in

$$D(1, 1), D(3, 1), D(3, 3), D(5, 3), D(5, 5), D(7, 5), D(7, 7), D(9, 7),$$

the complete Hamiltonian-cycle universe has size

$$|\mathcal{H}(D(a, b))| = ab + 5.$$

For each of the five state-encoded members from  $D(5, 3)$  through  $D(9, 7)$ , every cycle belongs to exactly one of three observed classes: one all-rails cycle,  $ab$  cycles determined by one turn cell in each ladder, and four two-connector exceptions.

Here is the local state mechanism behind the finite classification. Number the connectors in the order displayed in equation (3). A cycle crosses the four-edge cut between the two ladders an even, nonzero number of times. With all four connectors present, the degree-two equations along one ladder force the top and bottom rail decisions to agree cell by cell. In every certified universe, excluding an internal closed component leaves either the all-rails state or a single omitted rail pair bounded by two selected rungs. The latter is a turn at one of the ladder cells. Two straight ladders give the all-rails cycle; two turned ladders give one cycle for each independent pair of turn positions.

With exactly two connectors, the four observed Hamiltonian connector words are

$$1100, \quad 0011, \quad 1010, \quad 0101,$$

each with one forced completion for odd ladder lengths. The remaining words 1001 and 0110 do not occur in the certified complete universes.

**Theorem 6.2** (Hamiltonian cycles of the double ladder). *For odd  $a, b \geq 3$ , the Hamiltonian cycles of  $D(a, b)$  are one all-rails cycle, the  $ab$  cycles with one turn in each ladder, and four two-connector exceptions. In particular,*

$$|\mathcal{H}(D(a, b))| = ab + 5.$$

*Proof.* For a ladder of length  $m$ , put binary variables  $x_{t,s}$  on rail  $s$  of cell  $t$ ,  $y_t$  on rung  $t$ , and  $c_{0,s}, c_{m,s}$  on its endpoint connectors. At an internal vertex the degree-two equation is

$$x_{t-1,s} + x_{t,s} + y_t = 2,$$

with the analogous endpoint equations including  $c_{0,s}$  or  $c_{m,s}$ . Subtracting the equations on the two rails gives  $d_t = -d_{t-1}$ , where  $d_t = x_{t,0} - x_{t,1}$ . Thus the endpoint connector bits determine the alternating rail difference, after which every rung is forced.

If all four connectors are selected,  $d_0 = 0$ , so the two rail choices agree in every cell. A zero rail pair forces the two bounding rungs. Two such zero pairs enclose a separate cycle, so a connected spanning solution has either no zero pair or one. The two ladders must have the same type: one straight and one turned produces two components. Hence there is one all-rails cycle and one cycle for each independent pair of turn positions.

If exactly two connectors are selected, applying  $d_t = (-1)^t d_0$  at the far endpoint and using odd  $m$  leaves the four connector words 1100, 0011, 1010, 0101. In each case the recurrence has one completion: one ladder uses all rungs and alternating rails, while the other uses all rails and only the appropriate endpoint rung. The remaining words close a component before all vertices are reached. The connector cut must be crossed a positive even number of times, so zero, one, and three connectors cannot yield a Hamiltonian cycle. These cases exhaust the recurrence.  $\square$

## 7 Exact edge separation for double ladders

Throughout this section, let  $a, b \geq 3$  be odd. In ladder  $A$ , write  $r_A(p) = A_p^0 A_p^1$  for rung  $p$ ,  $0 \leq p \leq a$ , and  $\ell_A(i, s) = A_i^s A_{i+1}^s$  for rail  $s$  of cell  $i$ ,  $0 \leq i < a$ . Use  $r_B(q)$  and  $\ell_B(j, s)$  analogously. Denote the all-rails cycle by  $R$ , the cycle turning in cell  $i$  of  $A$  and cell  $j$  of  $B$  by  $T(i, j)$ , and the four exceptional cycles from theorem 6.2 by

$$X_{0011}, \quad X_{1100}, \quad X_{0101}, \quad X_{1010}.$$

The subscripts record connector words in the order of equation (3).

### 7.1 A symbolic packing lower bound

We first construct ordered edge requirements whose cycle supports are pairwise disjoint. Four requirements isolate the four exceptional cycles. One concrete choice, anchored on ladder  $A$ , is

$$(r_A(0), r_A(1)), \quad (r_A(a), r_A(a-1)), \quad (\ell_A(0, 0), \ell_A(0, 1)), \quad (\ell_A(0, 1), \ell_A(0, 0)). \quad (4)$$

The incidence recurrence in theorem 6.2 shows directly that the four supports are distinct singleton sets, one for each  $X$ -cycle.

Next isolate every boundary turn. For  $0 \leq j < b$ , use

$$(r_A(0), \ell_B(j, j \bmod 2)), \quad (r_A(a), \ell_B(j, 1 - j \bmod 2)). \quad (5)$$

They have singleton supports  $\{T(0, j)\}$  and  $\{T(a-1, j)\}$ , respectively. For  $1 \leq i \leq a-2$ , the symmetric requirements

$$(r_B(0), \ell_A(i, i \bmod 2)), \quad (r_B(b), \ell_A(i, 1 - i \bmod 2)) \quad (6)$$

have supports  $\{T(i, 0)\}$  and  $\{T(i, b-1)\}$ . The parity choice in equations (5) and (6) ensures that an exceptional cycle containing the endpoint rung also contains the excluded rail, so no exceptional cycle enters a boundary support.

It remains to handle the interior turn grid. For  $0 < p < a$ , theorem 6.2 gives

$$\text{supp}(r_A(p), \ell_B(j, s)) = \{T(p-1, j), T(p, j)\}, \quad (7)$$

and, symmetrically, for  $0 < q < b$ ,

$$\text{supp}(r_B(q), \ell_A(i, s)) = \{T(i, q-1), T(i, q)\}. \quad (8)$$

Here either rail side  $s \in \{0, 1\}$  may be used. These requirements are dominoes in the  $(a-2) \times (b-2)$  grid of interior turns.

Delete its far corner  $T(a-2, b-2)$ , which belongs to the majority parity class. In each interior  $B$ -column, pair the  $A$ -rows 1-2, 3-4,  $\dots$ ,  $(a-4)-(a-3)$ . In the remaining row  $a-2$ , pair the  $B$ -columns in the same way. This tiles every remaining interior cell and leaves precisely the deleted corner.

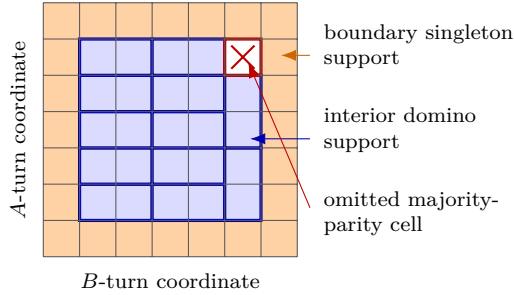


Figure 3: The packing construction, shown for a  $7 \times 7$  turn grid. Orange boundary cells and the four exceptional cycles have singleton support; blue rectangles are pairwise disjoint two-turn supports. The construction for general odd  $a, b$  is the same row-first domino tiling with one majority-parity interior cell omitted.

**Theorem 7.1** (Symbolic packing bound). *For odd  $a, b \geq 3$ ,*

$$\text{hsep}(D(a, b)) \geq \frac{(a+2)(b+2) - 1}{2}.$$

*Proof.* The four supports in equation (4), the boundary singleton supports, and the domino supports in equations (7) and (8) are mutually disjoint: they partition all Hamiltonian-cycle classes except  $R$  and the deleted interior turn. The all-rails cycle covers none of the chosen requirements. Thus every Hamiltonian cycle covers at most one chosen requirement. Their number is

$$4 + (2a + 2b - 4) + \frac{(a-2)(b-2) - 1}{2} = \frac{(a+2)(b+2) - 1}{2}.$$

The packing half of theorem 2.5 proves the claim.  $\square$

## 7.2 A symbolic separating-family upper bound

Define

$$\begin{aligned} \mathcal{C}(a, b) = & \{X_{0011}, X_{1100}, X_{0101}, X_{1010}\} \\ & \cup \{T(i, j) : i \in \{0, a-1\} \text{ or } j \in \{0, b-1\}\} \\ & \cup \{T(i, j) : 0 < i < a-1, 0 < j < b-1, i+j \text{ odd}\}. \end{aligned} \quad (9)$$

There are  $2a + 2b - 4$  boundary turns. The odd parity class of the odd by odd interior grid has  $((a-2)(b-2)-1)/2$  cells, and hence

$$|\mathcal{C}(a, b)| = 4 + (2a + 2b - 4) + \frac{(a-2)(b-2)-1}{2} = \frac{(a+2)(b+2)-1}{2}. \quad (10)$$

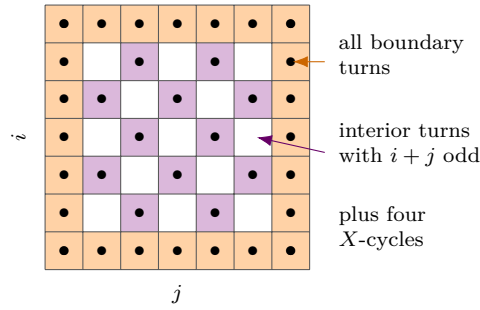


Figure 4: Turn cycles selected by  $\mathcal{C}(a, b)$ , again illustrated on a  $7 \times 7$  grid. Every boundary turn is selected, as is one checkerboard class of interior turns; the four connector-exception cycles are added separately.

We prove that this family separates by giving both directed witnesses for every pair of edge types. Let  $P$  be its selected turn cycles. Write  $R_i = \{T(i, j) \in P\}$ ,  $K_j = \{T(i, j) \in P\}$ , and

$$N_A(p) = \{p-1, p\} \cap \{0, \dots, a-1\}, \quad N_B(q) = \{q-1, q\} \cap \{0, \dots, b-1\}.$$

On the turn coordinates of an edge signature, every connector has all of  $P$ ; an  $A$ -rail in cell  $i$  has  $P \setminus R_i$ ; an  $A$ -rung  $p$  has precisely the selected turns whose first coordinate is in  $N_A(p)$ . The corresponding statements for ladder  $B$  use columns.

For the four exception coordinates, abbreviate  $B_0 = X_{0011}$ ,  $B_1 = X_{1100}$ ,  $A_0 = X_{0101}$ , and  $A_1 = X_{1010}$ . An  $A$ -rail contains  $B_0, B_1$  and exactly one of  $A_0, A_1$ , determined by its side and cell parity. An internal  $A$ -rung contains  $A_0, A_1$ ; an endpoint rung also contains one of  $B_0, B_1$ . Interchanging  $A$  and  $B$  gives the other ladder. In connector order equation (3), the four exception parts of the connector signatures are

$$\{B_1, A_1\}, \quad \{B_1, A_0\}, \quad \{B_0, A_1\}, \quad \{B_0, A_0\}. \quad (11)$$

**Theorem 7.2** (Symbolic separating family). *For odd  $a, b \geq 3$ , the edge signatures induced by  $\mathcal{C}(a, b)$  are pairwise incomparable. Consequently,*

$$\text{hsep}(D(a, b)) \leq \frac{(a+2)(b+2)-1}{2}.$$

*Proof.* We exhaust the eight edge-type combinations. In every case we identify a cycle in each of  $\sigma(e) \setminus \sigma(f)$  and  $\sigma(f) \setminus \sigma(e)$ .

1. *Two connectors.* The four sets in equation (11) are distinct and have size two. Each of two such sets therefore has an exception coordinate absent from the other.
2. *A connector and a rail.* A selected boundary turn in the rail's omitted row or column belongs to the connector but not the rail. A rail has three exception coordinates—both opposite-ladder exceptions and one alternating same-ladder exception—whereas a connector has only two; the unused opposite-ladder exception belongs to the rail alone.
3. *A connector and a rung.* Since  $a, b \geq 3$ , there is a boundary row or column outside the rung's one- or two-cell neighbourhood. A boundary turn there belongs only to the connector. The rung contains both same-ladder exceptions, while the connector contains exactly one of them, so the other exception gives the reverse witness.
4. *Two rails.* The two sides of one cell have the same turn coordinates but opposite alternating exception coordinates. Rails in different cells of one ladder are distinguished in both directions by selected boundary turns in their two omitted rows or columns. For rails on different ladders, choose a boundary point in the omitted column but outside the omitted row, and then one in the omitted row but outside the omitted column. Both are selected and give the two directions.
5. *Two rungs of one ladder.* If their neighbourhoods are not nested, a selected boundary turn in each set difference gives both witnesses. The only nested pairs are  $N_A(0) \subset N_A(1)$  and  $N_A(a) \subset N_A(a-1)$ , and their  $B$ -analogues. There the endpoint rung has an additional exception coordinate, while a turn in the internal rung's extra neighbouring cell gives the reverse witness.
6. *Rungs of different ladders.* Their two-element exception cores are  $\{A_0, A_1\}$  and  $\{B_0, B_1\}$ . An endpoint rung can add at most one coordinate from the other core, so each signature retains an exception absent from the other.
7. *A rail and a rung of the same ladder.* The rail contains both opposite-ladder exceptions and only one same-ladder exception. The rung contains both same-ladder exceptions and at most one opposite-ladder exception. Each therefore has an exception absent from the other.
8. *A rail of one ladder and a rung of the other.* Consider an  $A$ -rail in row  $i$  and a  $B$ -rung with neighbourhood  $N_B(q)$ . Choose a boundary row different from  $i$  and a boundary column outside  $N_B(q)$ . The corresponding selected turn belongs to the rail but not the rung. Conversely choose  $j \in N_B(q)$  so that  $T(i, j)$  is selected. If  $i$  is a boundary row, every  $j$  works. If  $i$  is interior, an endpoint neighbourhood supplies a boundary column, while an internal two-column neighbourhood contains exactly one column of the required parity. Then  $T(i, j)$  belongs to the rung but is omitted by the rail. The other orientation is symmetric.

The cases include endpoint rungs and both rail sides. A square insertion introduces only new rails and rungs with the same indexed incidences, so it introduces no additional edge type or boundary case. Thus all distinct edge signatures are incomparable. Apply theorem 2.4 and equation (10).  $\square$

### 7.3 The exact formula and its lift

**Theorem 7.3** (Exact double-ladder formula). *For all odd integers  $a, b \geq 3$ ,*

$$\boxed{\text{hsep}(D(a, b)) = \frac{(a+2)(b+2)-1}{2}}. \quad (12)$$

*Proof.* The lower bound is theorem 7.1; the upper bound is theorem 7.2. Their cardinalities agree.  $\square$

Both certificates lift locally under square insertion. For example, under  $b \mapsto b+2$ , keep column zero and shift every old positive column by two. The two new columns contribute four boundary cycles and one selected parity cycle in each of the  $a-2$  interior rows, for a net increase of  $a+2$ .

For the packing, the same strip contributes four boundary singletons and  $a - 2$  dominoes. The  $a \mapsto a + 2$  rule is symmetric and adds  $b + 2$ . At the level of graph edges, turns at the expanded cell and two alternating exceptions require the uniform local replacement inside the new square gadget; all other cycles lift by replacing subdivided rails with their three-edge paths.

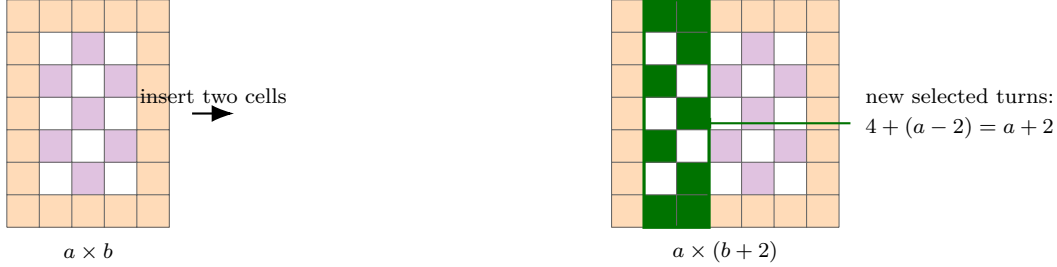


Figure 5: The separating-family lift for  $b \mapsto b + 2$ , illustrated with  $a = 7$ . Green cells are precisely the new selected turns: four boundary turns and one parity turn in each interior row. The packing lift has the same increment, with the interior additions represented by domino supports.

The theorem concerns  $D(a, b)$  itself. It does not assert that these graphs maximize hsep among all Barnette graphs of their orders.

## 8 Prospective confirmations beyond order 36

The double-ladder formulas were tested prospectively before their general proofs were completed. A timestamped, read-only prediction record froze the following values before any of the three target graphs was constructed or analyzed:

| $G$         | $ \mathcal{H}(G) $ | $\text{hsep}(G)$ |
|-------------|--------------------|------------------|
| $D(9, 9)$   | 86                 | 60               |
| $D(11, 9)$  | 104                | 71               |
| $D(11, 11)$ | 126                | 84.              |

**Computational Result 8.1** (Prediction-locked double-ladder test). All six frozen values were subsequently confirmed exactly. Each graph was obtained from the certified  $D(9, 7)$  by the specified facial square insertions. For each target, a path-based enumerator and an independent perfect-matching/complement enumerator produced identical complete normalized Hamiltonian-cycle universes. Explicit primal and matching packing certificates established the three displayed hsep values, and the standard-library verifier passed every graph, cycle, ordered edge-pair, lower certificate, and manifest check.

Historically, these were genuinely prospective checks of both formulas. Mathematically, the Hamiltonian-cycle values now also follow from theorem 6.2, and the hsep values from theorem 7.3. The computations remain useful independent finite checks of the symbolic descriptions and of three square-insertion lifts.

No complete Barnette census was run at order 40, 44, or 48 for this test. Accordingly, Computational Result 8.1 proves neither  $M_B(40) = 60$  nor any value of  $M_B(44)$  or  $M_B(48)$ ; it makes no order-level extremal claim.

## 9 Tied orders and open extremal conjectures

### 9.1 Tied maximizers in the certified range

The extremal behavior at orders congruent to two modulo four is structurally different from the uniform double-ladder recurrence.

**Computational Result 9.1** (Certified tied-order structure). Every one of the 17 tied maximizers at orders 22, 26, 30, 34 contains a 3-cycle component in the face-adjacency graph induced by quadrilateral faces, as well as at least one strict ladder component. The full quadrilateral component signatures are pairwise distinct within each order and reproduce the tie counts 2, 5, 3, 7.

Twenty-five exact square-gadget certificates link these graphs to earlier examples. Thirteen links are literal ladder-cell insertions. In the other twelve, the same four-vertex gadget is inserted on two suitable disjoint edges of a 6-, 8-, or 10-face end-cap region. Thus these graphs are not asserted to be members of one  $D(a, b)$  recurrence.

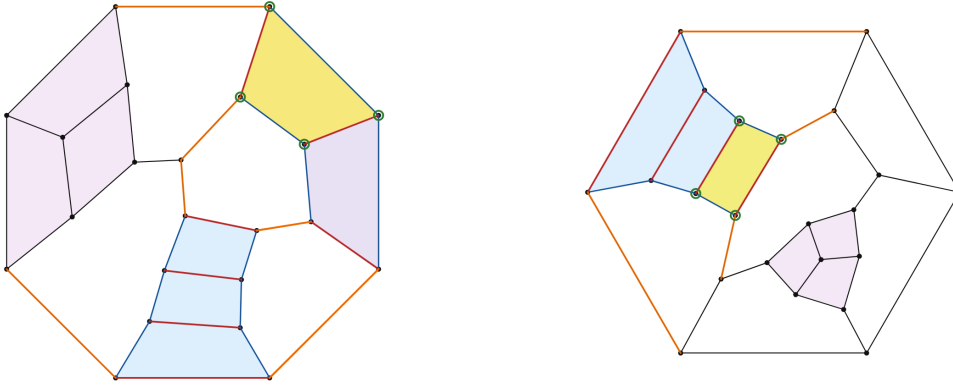


Figure 6: The two certified order-22 maximizers. Purple marks the quadrilateral 3-cycle component; blue and red distinguish ladder rails and rungs. Their different component signatures illustrate why the tied branch is not identified with a single double-ladder skeleton.

### 9.2 Open extremal continuation

For each order divisible by four through 36, the unique certified maximizer is the balanced or nearly balanced double ladder listed in Computational Result 5.4. The exact value of  $\text{hsep}(D(a, b))$  is now known for all odd  $a, b \geq 3$ , but this does not compare  $D(a, b)$  with all other Barnette graphs at the same order.

**Conjecture 9.2** (Extremal continuation). *At orders divisible by four beyond the certified census range, the balanced or nearly balanced double ladder is the unique Barnette graph maximizing Hamiltonian edge separation.*

Proving Conjecture 9.2 requires order-level arguments or complete competing-graph certificates; the exact family formula equation (12) alone is insufficient. In particular, this manuscript does not

claim  $M_B(40) = 60$ . The previously recorded linear fit to the three tied-order values at 26, 30, 34, which predicts  $M_B(38) = 39$ , remains an especially weak extrapolation rather than a conjecture supported by a transfer proof. No order-38 computation is reported here.

## 10 Reproducibility and independent verification

The source repository contains the producer, structural analysis, tests, and standalone-verifier source. Large immutable result packages remain outside the repository. The manuscript itself contains only the compact table and selected figure files needed to compile it.

The global standard-library verifier checked all 29 order packages from 8 through 36, including the 14 nonempty orders, 266 manifest-listed files, every cycle certificate, every ordered edge-pair cover, all lower certificates, and the global summaries. Its recorded runtime was 655.989 seconds. The producer runtimes recorded across the orders sum to 5,186.661 seconds; those values mix fresh production and the imported order-36 workflow and are provided as audit metadata rather than a hardware-independent benchmark.

The principal immutable identifiers used for this draft are

| package                        | SHA-256 of SHA256SUMS.txt                                        |
|--------------------------------|------------------------------------------------------------------|
| certified sequence             | 2a0a27cbb2c5dcefca3173293dc1fbfb21b55a69cdd366da3ab3b30a2842affa |
| order-36 subpackage            | 6aaa43a275ecbbedd25586fc244140ddfb7f107e14025b2bfd1da34af417d00d |
| ladder analysis                | 17b1a8219321fc70ac8146f17e29d8b836c90862ec96248429141869e252a67e |
| prospective double-ladder test | 6bd843dfbe9f3eb1aa3213511ce28091efc7567f7d7565f0aa980943b05713a1 |
| packing-lift output            | 188800bd2a8b0835216f6fa9c78eb4673e5b3bc24d1dc216bc3a7d619986f31e |
| primal-lift output             | bd5fdc3e3a160b8339fa412d4c1decf2113919cf1f4f6c2dc8856b474aeb37d1 |

The six manifests contain respectively 266, 9, 30, 50, 25, and 13 entries. Every listed file digest was rechecked before this revision, without modifying any package. The packing- and primal-lift source inventories bind their outputs to the order-36, ladder-analysis, and prospective source artifacts. Their recorded independent-verification results are **pass**; the primal package verifies 2,862, 3,540, 4,290, and 5,112 ordered edge requirements for the four finite double ladders.

The current repository test run collected 191 tests: 186 passed and five environment-gated external integrations were skipped. The skipped tests require explicitly configured plantri or cubhamg executables and do not replace the standalone certificate verification described above.

The prospective package locks its predictions in a read-only JSON record with SHA-256 **e8ba381dd163c59941db81f65b71b70d6d78cdb7cfd6f6a41ba4828fe25c89fb**. Its standalone verifier passed all three constructions, complete universes, structural classifications, primal covers, matching lower certificates, and local and global manifests.

The planar layouts in this manuscript derive from certified rotation systems using Gunnar Brinkmann’s external drawing program. The wrapper checked that the vertices and edges recovered from every generated drawing exactly match the corresponding certified graph. Use of that program does not imply endorsement, collaboration, or coauthorship.

### Draft AI-assistance disclosure

AI tools assisted with software development, data analysis, drafting, and language editing. All mathematical certificate packages are independently checkable. The author remains responsible for the definitions, claims, proofs, source verification, interpretation of computational evidence, and final manuscript.

## 11 Conclusion

The certified computation determines  $M_B(n)$  for every nonempty Barnette census from order 8 through order 36, including all ties and the unique order-36 value 49. Its proof objects are explicit: upper certificates are Hamiltonian-cycle families, and lower certificates are either packings or complete-universe exhaustive exclusions checked by a standalone verifier.

That computation exposed a structural double-ladder recurrence. A rail/rung recurrence now proves that  $D(a, b)$ , for odd  $a, b \geq 3$ , has one all-rails cycle,  $ab$  turn cycles, and four connector exceptions. The resulting coordinate system supports two matching symbolic constructions. Exception and boundary singletons together with an interior domino tiling form a packing of  $((a + 2)(b + 2) - 1)/2$  requirements. The four exceptions, all boundary turns, and one interior parity class form a separating family of the same size; its rail, rung, and connector signatures are pairwise incomparable. Thus the exact formula

$$\text{hsep}(D(a, b)) = \frac{(a + 2)(b + 2) - 1}{2}$$

holds throughout this infinite family.

The three prediction-locked graphs beyond order 36 provide prospective finite confirmation of the cycle and edge-separation formulas, but no census-level claim at orders 40, 44, or 48. Likewise, the exact family theorem does not prove that balanced double ladders remain extremal among all Barnette graphs. That continuation, and a structural explanation of the tied boundary-state branch at orders congruent to two modulo four, remain open.

Before circulation as a novelty claim, the manuscript still requires a systematic literature review of related separation systems, cycle-separation parameters, Barnette censuses, and graph-expansion operations. Independent expert review of the symbolic case analysis and the direct 3-connectivity argument also remains desirable.

## References

Gunnar Brinkmann, Brendan McKay, and Heidi Van den Camp. *Guide to Using plantri (Version 5.8)*, n.d. URL <https://users.cecs.anu.edu.au/~bdm/plantri/>. Software guide distributed with plantri 5.8.